



How may I help you? Driving brand engagement through the warmth of an initial chatbot message

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ABSTRACT

Despite the growing number of brands that rely on chatbots to address customer service inquiries that once required human intervention, academics and practitioners are only beginning to acknowledge the role of chatbots in brand-building activities. Chatbots can initiate online conversations, thereby often serving as a consumer's first brand impression. However, little is known about how managers can strategically tailor a chatbot's initial message to foster consumer–brand connections and, ultimately, engagement. Three studies demonstrate that when chatbots initiate a conversation using a warm (vs. competent) message, brand engagement increases, as assessed using both computerized text analysis and traditional scale measures. Brand–self distance mediates this effect, such that a warm (vs. competent) initial chatbot message makes consumers feel closer to the brand. Further, the authors identify brand affiliation as a theoretically relevant moderator. This research thus offers managers insight into how initial chatbot messages can attract and engage consumers.

1. Introduction

Technological innovations have been instrumental in offering brands the potential to reduce costs and enhance productivity in customer service contexts (Canhoto & Clear, 2020). One growing way in which brands use technology in service interactions is through the adoption of online conversational agents, such as chatbots (Murtarelli et al., 2021). Traditionally, chatbots were used for low-involvement customer service tasks, such as booking reservations or providing more detailed product information (Davenport & Ronanki, 2018; Joshi, 2018; Wagner, 2016). However, technological advances mean that today chatbots can provide answers to about 80% of consumer questions (Jovic, 2020; Verma, 2019). Chatbots offer an opportunity to provide consumers with a “consistent and enhanced” service experience in which the conversational agent embodies the brand (Köhler et al., 2011, p. 93)—and brands appear to be capitalizing on this opportunity. The chatbot market is currently worth nearly \$3 billion and is projected to exceed \$9 billion by 2024 (Nguyen, 2020). By 2021, about 85% of customer interactions will be handled without the intervention of a human agent (Jovic, 2020).

Artificial intelligence (AI) is revolutionizing the traditional paradigm of service exchanges, with software constantly improving in terms of its ability to mimic human-to-human interactions (Libai et al., 2020). Still,

more studies are needed to identify best practices and guidelines related to the use of chatbots in service and brand-building contexts (Holtgraves et al., 2007; Robinson et al., 2020). In particular, further research is warranted to understand how technology can be leveraged to foster personal connections with consumers (Rafaeli et al., 2017).

When used strategically, chatbots represent an opportunity for brands to effectively and efficiently interact with consumers. However, it is important for managers to acknowledge that chatbots also have shortcomings when it comes to generating consumer engagement. Research shows that consumers perceive chatbots to lack empathy and consideration of personal feelings (Luo et al., 2019). Additionally, consumers reject chatbots if they perceive them to be an interruption or if the conversation feels too impersonal (Computer Generated Solutions, 2019). Consumers also become frustrated with chatbots if they are unable to resolve complicated concerns (Trivedi, 2019). And while consumers are more likely to reject error-prone chatbots, chatbots that seek clarification to avoid potential miscommunication do not reduce likelihood of acceptance or adoption relative to error-free chatbots (Sheehan et al., 2020).

Despite the widespread use of chatbots in the industry, there have only been a few studies on this topic in the marketing literature (e.g., Chung et al., 2020; Marinova et al., 2017; Sands et al., 2021; Sheehan et al., 2020). One of the major takeaways of this limited research is that

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interactions with AI technologies implicitly carry a social component—whether the interaction is between a firm employee and an AI system (Makarius et al., 2020) or a consumer and an AI agent (Holtgraves et al., 2007). For instance, prior research suggests that consumers expect chatbot interactions to be characterized by friendliness while also offering informational value (Köhler et al., 2011; Shum et al., 2018). Although one of the benefits of chatbots is the ability to provide relatively homogeneous and predictable customer service, this can also detract from the emotional and social value of consumer–brand interactions. Specifically, Sands et al. (2021) find that customer service script content differentially affects satisfaction and purchase intention, such that when an educational script is used, human agents are more effective than chatbots, but when an entertaining script is used, no differences exist between the two modes of service delivery. We contribute to this emerging stream of work by investigating how varied approaches to initiating chatbot conversations might influence consumer–brand relations. In doing so, we answer the call for research to consider how chatbots can foster two-way communication and engagement through a service encounter between a brand and a consumer (Kim et al., 2021). Though both warmth and competence are valued by consumers, little is known about how brands should formulate initial chatbot messages in order to encourage brand engagement.

Against this backdrop, the present research explores how the initial message of a chatbot can affect consumer engagement. Studying the implications of a chatbot's initial message is important for both theory and practice. Such an investigation promises to enrich social response theory (Holzwarth et al., 2006; Köhler et al., 2011) by examining the potential role of chatbot's initial message as a social cue that may foster consumer likelihood to respond. By exploring whether and how a warm (vs. competent) initial message can shape instant consumer impressions and behaviors, our work also enhances research in social psychology that has defined warmth and competence as two fundamental dimensions of social perception (Fiske et al., 2002). Such insights carry significant managerial value as well, considering that the capability to engage consumers socially has become a critical success factor for frontline technology in general (van Doorn et al., 2017) and chatbots in particular (e.g., Phillips, 2018; Steele, 2018). Reflecting the growing challenge to generate leads in today's cluttered marketing and media landscape, U.S. advertising spending on digital lead generation is forecast to grow to \$3.2 billion in 2023 (Statista Research Department, 2021). Thus, studying how AI tools, such as chatbots, may help with the generation of leads is becoming essential (Kumar et al., 2021).

Consistent with research on social categorization and stereotyping (Allport, 1954; Fiske et al., 2002; Macrae et al., 1994), we propose and demonstrate across three experiments that a chatbot which uses a warm (vs. competent) initial message increases brand engagement. Furthermore, we postulate and show that a warm (vs. competent) initial chatbot message reduces brand–self distance, which explains our proposed warmth–engagement effect. Finally, we identify the moderating role of brand affiliation. In line with our brand–self distance explanation, we find that a chatbot's warm (vs. competent) initial message is critical when consumers already feel connected to the brand. Our work shows the effect of message warmth on brand engagement using different types of measurements, including a behavioral engagement proxy in which consumers provide a real response to a chatbot message. The results of our studies provide empirical evidence that can guide managers on how to create chatbot messages—specifically during the initial stages of an online interaction when the risk of consumer drop-off from websites is greatest (Chan, 2019).

2. Conceptual framework and hypotheses

2.1. Chatbots and marketing

AI is a combination of technological elements that complete tasks that would traditionally require the intervention of human intelligence

(Kumar et al., 2021; Martínez-López & Casillas, 2013). Quickly becoming a solution for business initiatives once thought to be solely reserved for humans (Campbell et al., 2020), AI offers brands the opportunity to enhance productivity and reduce costs (Canhoto & Clear, 2020). Among the fastest-growing AI technologies are chatbots, which are software programs that use natural language to engage in real-time conversations with consumers (Holtgraves et al., 2007; Joshi, 2018). Chatbots are most commonly implemented in customer service contexts and therefore interact directly with consumers (Brandtzaeg & Følstad, 2018; Chung et al., 2020).¹ Industry reports suggest that chatbots can cut a firm's operational costs by up to 30% (Jovic, 2020) while simultaneously benefiting consumers by enabling them to access seamless and monitored customer support around the clock (Vora, 2018).

Though the number of brands that use chatbots to interact with consumers continues to grow, there is a dearth of research dedicated to exploring the marketing-related implications of AI (Martínez-López & Casillas, 2013). Current knowledge is also limited about how AI tools, such as chatbots, can be harnessed to integrate overall marketing strategies (Kumar et al., 2021), improve customer experience, and enhance customer–brand relationships (Köhler et al., 2011). One of few notable exceptions to this lack of marketing research is the finding that consumers tend to evaluate a chatbot's personality as they do that of a human they meet for the first time, such that they can differentiate chatbot personalities after only a brief interaction (Holtgraves et al., 2007). Similarly, empathy (i.e., the ability to recognize, understand, and respond compassionately to the emotion of another; Levenson & Ruef, 1992) improves consumer receptiveness to AI agents, such as chatbots (Fung et al., 2018; Huang & Rust, 2018; Leite et al., 2013).

The current research expands this early body of work on how social cues (e.g., personality, empathy) can affect consumer reactions to AI agents by examining whether and how the tone of an initial chatbot message alone can serve as a social cue and, in turn, affect brand engagement, including consumers' likelihood to respond to the message and their motivation to learn more about the brand on whose behalf the chatbot operates. While prior research suggests that consumers value chatbots for serving as not only information sources but also virtual companions (Shum et al., 2018), it has not yet been systematically studied whether an initial chatbot message that conveys warmth or competence can have a differential effect on brand engagement. Studying the initial interaction is critical given the growing importance of chatbots for brands to the extent that if a brand's chatbot makes a negative first impression, consumers may elect to never again patronize the brand (EPAM, 2017).

2.2. Chatbots and the stereotype content model

Considering that consumers tend to quickly draw inferences about a chatbot and its personality in similar ways in which they do about another person they first encounter (Holtgraves et al., 2007), we build our conceptual framework on consumer reactions to the warm or competent tone of a chatbot's initial message by drawing on research on social categorization (i.e., the effortless and spontaneous process of grouping people into social groups; Allport, 1954) and stereotyping (i.e., the process of simplifying a complex social environment by activating and applying social category information; Macrae et al., 1994). One of the most established frameworks to analyze social stereotyping is the Stereotype Content Model (SCM; Fiske et al., 2002). Indicating the importance of first impressions in interpersonal encounters, the SCM

¹ Chatbots are distinct from virtual assistants (VAs). Although both chatbots and VAs are conversational agents, VAs (also referred to as intelligent personal assistants; Shum et al., 2018), such as Amazon's Alexa and Apple's Siri, are designed primarily to serve as assistants that handle individual-specific tasks, such as setting reminders, checking the weather, and scheduling meetings (Gnewuch et al., 2017; Joshi, 2018; Misal, 2018).

suggests that there are two universal dimensions of social cognition: warmth and competence. Evolutionary pressures explain why people who meet someone for the first time tend to try to assess as quickly as possible whether the individual has good intentions (i.e., is a warm person) and whether the individual can act upon these intentions (i.e., is a competent person). Warmth perceptions capture traits such as friendly, generous, honest, righteous, sincere, and tolerant. Competence perceptions encompass traits such as creative, efficient, intelligent, and knowledgeable. These two universal dimensions of social cognition can explain interpersonal as well as intergroup relationships (Fiske et al., 2007).

Propelled by the increasingly accepted notions that people relate to brands similarly to how they relate to other people (Fournier, 1998), that consumers tend to attribute humanlike characteristics, such as emotions, intentions, and motivations, to nonhuman entities, such as brands (Aggarwal & McGill, 2007; Hur et al., 2015), and that consumers care not only about a brand's ability to deliver on its claims but also about the intentions underlying its actions (MacInnis et al., 2009), researchers started exploring warmth and competence perceptions in a branding context. For example, Aaker et al. (2010) have found that consumers tend to engage in brand stereotyping by perceiving nonprofit companies as warmer than for-profit companies. More recently, proposing that consumer–brand relationships are driven by the same two universal dimensions of warmth and competence, Kervyn et al. (2012) have adapted the SCM to fit brand perception by developing the Brands as Intentional Agents Framework (BIAF). In the BIAF, warmth stands for a brand's perceived intention, whereas competence stands for its perceived ability. Both perceived brand intent and perceived brand ability have been found to independently predict favorable consumer responses, such as purchase intentions and brand loyalty (Kervyn et al., 2012).

Little is known, however, about whether the initial display of warmth or competence in an online environment, such as how a chatbot initiates a conversation with consumers on behalf of a brand, differentially impacts consumers' immediate brand reactions and, in turn, their likelihood to engage with the brand further. According to research in social psychology, initial impressions directly influence subsequent expectations and behavior (Jones, 1990), so the impact of a chatbot's warm (vs. competent) initial message on brand engagement is important to examine—not only theoretically, because it would further expand the scope of the SCM and demonstrate its broad appeal and relevance, but also practically, because a critical success factor of a chatbot as an interactive agent is its instant ability to engage consumers in a conversation (Phillips, 2018; Steele, 2018).

2.3. Warm versus competent chatbots and consumer engagement

Over the past one and a half decades, consumer engagement has emerged as one of the most central marketing concepts. After years of debate, there appears to be growing consensus in the literature on the multidimensional nature of the consumer engagement concept. In a nutshell, consumer engagement represents a consumer's motivational and context-dependent psychological state elicited by the consumer's varying degrees of affective, cognitive, and behavioral investments in interactions with a focal object (e.g., a brand). As such, compared with other relational concepts, such as involvement and participation, engagement captures the notion of co-creative, interactive experiences far more comprehensively (e.g., Brodie et al., 2011; Hollebeek, 2007, 2011; van Doorn et al., 2010). High levels of consumer engagement are important for brands to achieve, as they have been found to predict stronger consumer–brand relationships by increasing consumers' brand attachment, commitment, trust, loyalty, and other favorable relational and behavioral outcomes (e.g., Brodie et al., 2013; Kull & Heath, 2016).

In brands' growing battle for consumer attention, AI-infused frontline encounters through the use of chatbots provide promising opportunities to elicit consumer engagement (Shum et al., 2018). Though

initial work suggests that empathy and humanlike cues help AI agents, such as chatbots, evoke more favorable consumer reactions (Araujo, 2018; Huang & Rust, 2018), to the best of our knowledge, no prior research has systematically examined how a chatbot's initial message that comes across as either primarily warm or primarily competent may impact consumer motivation to respond. The accepted comprehensiveness of warmth and competence as the two central dimensions of social perception (Fiske et al., 2007) points to the importance of investigating the consequences of the perceived intent and ability of a brand's virtual service agent (i.e., chatbot), particularly given that a chatbot message often represents one of the first impressions a brand makes on a consumer and is often even the first direct brand interaction with a consumer (Banfi, 2017; Franzen, 2020).

While both warmth and competence are critical to social perception, research on general impression formation suggests that perceived warmth elicits stronger affective and behavioral responses than perceived competence, as warmth-related traits tend to be more accessible, predictive, and heavily weighted in evaluations than competence-related traits (Wojciszke et al., 1998). The primacy of warmth appears to be driven by individuals' greater cognitive sensitivity to warm (vs. competent) information, which enables them to reliably arrive at moral–social judgments more rapidly than at competence–efficacy judgments. This prevailing influence of warmth has been found to hold in lexical decision tasks, in which individuals across different age groups were able to identify warmth-related traits more quickly than competence-related traits (Ybarra et al., 2001), and in 100-ms exposures to faces, which led to more reliable first impressions of trustworthiness than competence (Willis & Todorov, 2006).

In keeping with these findings in social psychology, recent work in the branding literature also suggests that a brand's perceived intentions (i.e., warmth) tend to elicit more pronounced and predictive responses among consumers than the brand's perceived ability (i.e., competence). For example, consumers evaluated a brand that contributed to a disaster (e.g., economic crisis, oil spill) more stringently when the brand's intentions rather than its ability were called into question (Kervyn et al., 2014). On the upside, unlike brand competence perceptions, brand warmth perceptions have been found to increase consumer intentions to endorse a brand on social media (Bernritter et al., 2016).

Warmth and competence have also been studied in service settings, in which warmth is largely associated with a provider's customer focus, friendliness, and sociability, whereas competence is largely associated with a provider's confidence, efficiency, and knowledge (Grönroos, 1990; Kirmani et al., 2017; Tsai & Huang, 2002). In traditional service encounters, warmth has been associated with positive facial expressions, positive gestures and moving toward customers, and positive communication (Lemmink & Mattsson, 1998). Smile intensity has been found to have a differential effect on warmth and competence, such that consumers are more likely to perceive a broad (vs. slight) smile as warmer but less competent (Wang et al., 2017). Though most physical or vocal attributes used to display warmth and competence are restricted to in-person interactions, a novel stream of research is emerging that studies alternative ways of how warmth and competence as the two universal social dimensions can be evoked in online service contexts. Initial work in this area indicates that consumers tend to perceive service agents who use (vs. do not use) emoticons in digital business communications as warmer but less competent (Li et al., 2019). Warm (vs. competent) messages used by retailers in response to consumer complaints on online platforms are more likely to improve service perceptions of nonparticipating observers and increase positive word-of-mouth intentions (Huang & Ha, 2020).

Taken together, these findings across various fields of research converge on the notion that warmth perceptions tend to be more central than competence in determining approach–avoidance tendencies (Fiske et al., 2007), which suggests that a warm (vs. competent) initial chatbot message should make it more likely for consumers to respond and should increase consumers' motivation to learn more about the chatbot's brand.

Given the relational dimension of warmth (e.g., Fiske et al., 2002), this increased consumer motivation to engage is likely to be rooted in how close a warm (vs. competent) initial chatbot message makes the consumer feel toward the brand. The next section further explores this potential mechanism underlying the proposed chatbot warmth–consumer engagement effect.

2.4. The role of brand–self distance

The impact of warmth on consumer–brand relationships has found empirical support in the branding literature. For example, a panel survey of German household consumers has found that brand warmth predicts consumer–brand identification, which in turn predicts brand loyalty and advocacy (Stokburger-Sauer et al., 2012). A more recent study has confirmed brand warmth as a driver of consumer–brand identification and the latter as a predictor of purchase likelihood and brand ownership, whereas brand competence has been found not to affect consumer–brand identification (Kolbl et al., 2019). Both studies indicate that consumers are more likely to identify with a brand that they consider high in warmth, which is consistent with the notion that warmer brands contain more identity-related meanings or values (Kervyn et al., 2012; Stokburger-Sauer et al., 2012).

An essential component of consumer–brand identification is consumers' perceived distance between the identity of a brand and their own identity (Currás-Pérez et al., 2009). The more the identities are perceived to overlap, the more consumers identify with the brand and the closer they feel to it. This perceived degree of closeness has been conceptualized as brand–self distance, which is the extent to which consumers feel close (i.e., connected) to or distant to (i.e., disconnected from) a brand (Park et al., 2013). Building on the emerging literature on the effects of brand warmth on consumer–brand relationships (e.g., Kolbl et al., 2019; Stokburger-Sauer et al., 2012), we therefore posit that a chatbot's warm (vs. competent) initial message will make the recipient (i.e., consumer) feel more connected (i.e., closer) to the brand on whose behalf the chatbot operates. In addition, prior research has shown that consumers who feel closer to a brand exhibit higher brand involvement (Park & Mittal, 1985) and are more likely to engage with the brand by participating in co-creation campaigns (Hsieh & Chang, 2016).

In sum, we predict that a chatbot's warm (vs. competent) initial message should decrease consumers' perceived distance to the chatbot's brand, which in turn should increase the consumer's likelihood to engage with the brand further—not only affectively or cognitively by feeling more engaged with the brand or being more motivated to learn more about the brand and engage with it further but also behaviorally by being more likely to reciprocate the chatbot's initial message and thus respond to it. Formally, we therefore hypothesize the following:

H1. A chatbot's warm (vs. competent) initial message increases brand engagement.

H2. Brand–self distance mediates the effect of a chatbot's warm (vs. competent) initial message on brand engagement, such that a chatbot's warm (vs. competent) initial message makes consumers feel closer to the brand on whose behalf the chatbot operates, which in turn increases their engagement with the brand.

2.5. The role of brand affiliation

We propose that a warm (vs. competent) initial chatbot message reduces brand–self distance. If reading a warm (vs. competent) chatbot message enhances perceived closeness between the consumer and the brand, then brand affiliation should play an important role in how consumers react to a brand's chatbot. The branding literature has repeatedly shown that social expectations that occur in interpersonal contexts often replicate in consumer–brand relations (e.g., Aggarwal, 2004; Choi & Winterich, 2013; Fournier, 1998).

One important aspect that can affect consumer–brand relationships

is communication style (Johnen & Schnittka, 2019). Indeed, research shows that informal (vs. formal) communication styles in social media can backfire and lead to lower brand trust when consumers feel distant from the brand (Gretry et al., 2017). This is explained by compliance with social norms, such that consumers do not expect informal language (e.g., “awesome”) when they feel a separation between themselves and the brand.

In our work, we explore how the level of warmth in a chatbot message can affect brand engagement. Given that language or tonality expectations differ depending on the consumers' preexisting relationship with the brand, we expect that brand affiliation will moderate our effects. Specifically, affiliation in a marketing context is an indicator of the strength of the social bond between consumers and a brand (Paulssen et al., 2014). Therefore, when consumers are affiliated with a brand, they anticipate that the brand as a social partner will be warm, as this is an indicator of a high-quality social partner (Russell & Fiske, 2008). Consumers expect relationships with a brand they are affiliated with to be characterized by closeness and prefer the brand to use language that reflects the nature of such relationships (Sela et al., 2012). As such, warm messages, which are related to friendliness and sociability (Kirmani et al., 2017; Li et al., 2019), should be particularly well received in the context of a close relationship. Conversely, a lack of brand affiliation should create a distance between the consumer and the brand (Sela et al., 2012). Furthermore, explicitly emphasizing the absence of an affiliation highlights to consumers that they do not share a social bond with the brand, which should cause them to classify the brand as an outgroup (i.e., a brand that they do not feel they belong to; Wang et al., 2018). Accordingly, consumers would not expect such a brand to communicate with warm messages (Sela et al., 2012), so the use of a warm (vs. competent) message should not prove beneficial for creating consumer engagement.

Considering that consumers expect brands to act in a manner that is consistent with social norms (Aggarwal, 2004), we postulate that highlighting consumers' affiliation (lack of affiliation) with a given brand will reduce (enhance) brand–self distance and, in turn, change how consumers react to the initial chatbot message. In particular, we predict that when we highlight a close relationship with a brand, a chatbot's warm initial message—consistent with the language that consumers expect from a brand that is close to them—will create stronger brand engagement than a competent message. However, when we highlight a distant relationship, the warmth (vs. competence) of a chatbot message should not make a difference.

H3. Brand affiliation moderates the effect of a chatbot's warm (vs. competent) initial message on brand engagement.

3. Overview of studies

Three experiments test our conceptual framework and hypotheses depicted in Fig. 1. Study 1 uses a text analysis tool to investigate how consumers respond to warm (vs. competent) chatbot messages. In this study, we are able to examine consumers' behavioral engagement. Study 2 provides additional support for H1 regarding the effect of warm (vs. competent) chatbot messages on brand engagement, while also examining the underlying process through brand–self distance (H2). Study 3 provides further process evidence by showing how creating distance between consumers and the brand through contextual cues (i.e., a pre-existing brand affiliation or nonaffiliation) can moderate the effect (H3).

4. Study 1

We argue that a warm (vs. competent) message increases consumer engagement with a brand that uses AI in the form of a chatbot. In this first study, we will demonstrate this effect by simulating an initial interaction that would occur via text messaging on a smartphone. That is, participants will see either a warm or a competent text message sent

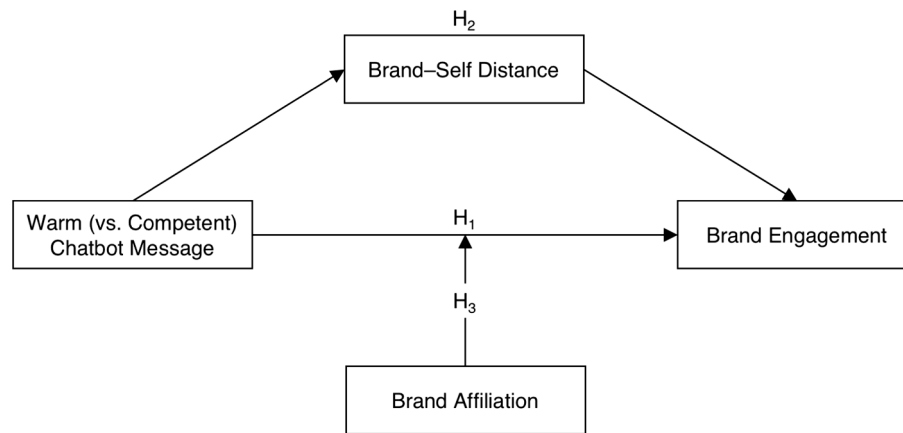


Fig. 1. Conceptual framework. A warm (vs. competent) initial chatbot message increases brand engagement by decreasing perceived brand–self distance. The effect of a warm (vs. competent) initial chatbot message on brand engagement is accentuated when consumers are affiliated with the brand.

to them by a brand and will be given the opportunity to provide a response. In this way, we can simulate behavioral engagement with the brand (van Doorn et al., 2010).

4.1. Method

4.1.1. Pretest

We pretested the stimuli used to examine the proposed effects. For Studies 1 and 2, we used chatbots that operated in a mobile setting. As such, we created initial messages and displayed them on a plain white screen that was designed to resemble that of a smartphone. The initial message of the chatbot—purportedly sent on behalf of a fictitious travel brand that we called Viatti—was primarily warm or primarily competent. The chatbot’s warm message read as follows: “Greetings! This is your personal travel guide by Viatti. I am so happy to welcome you. I will take care of you and try answering any travel questions you might have. How may I help you?” The chatbot’s competent message read as follows: “This is the virtual travel expert by Viatti. Years of experience in the travel industry enable me to answer any travel question. Let me know what travel information I should provide.” For a control condition we included in Study 2, we added a neutral chatbot message, which read as follows: “This is the messenger by Viatti.”

Eighty Amazon Mechanical Turk (MTurk) participants ($M_{\text{age}} = 40.41$ years; 50% female) were randomly assigned to one of the three text messages sent to them by the chatbot of the travel brand Viatti to initiate a conversation.² They were then asked to indicate to what extent they agreed or disagreed with each of the following two statements: “The initial message comes across as warm and friendly,” and “The initial message comes across as competent and capable” (1 = strongly disagree, 7 = strongly agree; Kervyn et al., 2012). All three manipulations worked as intended regarding “warm and friendly” ($M_{\text{warm}} = 5.62$, $M_{\text{competent}} = 4.88$, and $M_{\text{neutral}} = 2.86$; all p s < .05) as well as “competent and capable” ($M_{\text{warm}} = 5.46$, $M_{\text{competent}} = 6.20$, and $M_{\text{neutral}} = 4.45$; all p s < .05).

4.1.2. Participants, design, and procedure

Eighty-two students ($M_{\text{age}} = 20.71$ years; 45% female) from a large U.S. university participated in this study. The study used a single-factor (chatbot message: warm vs. competent) between-subjects design. Participants were instructed that they were planning to travel soon and that

a brand, Viatti, initiated a conversation with them. They were then randomly assigned to either the warm or the competent version of the pretested message stimuli. Thereafter, participants were asked to type a message in response to Viatti. Lastly, they provided some general demographic information (i.e., age and gender). Controlling for age and gender by including them as covariates did not alter any of the effects reported across all studies.

4.2. Results

For Study 1’s primary purpose of exploring consumers’ initial responses to chatbot messages that differ in their warm (vs. competent) tone, we conducted an automated text analysis with the support of the Linguistic Inquiry and Word Count (LIWC) software program (Pennebaker, Booth, et al., 2015). The LIWC program operates by counting the number of words that belong to a given category (Hartmann et al., 2019; Pennebaker, Boyd, et al., 2015). To operationalize and measure, in particular, the behavioral dimension of consumer engagement (van Doorn et al., 2010), we were interested in the total number of words that consumers used to respond to Viatti’s message. Three participants were excluded from the analyses, as they did not respond to the chatbot but instead directed their reply to the researchers, indicating that they would not provide a response (e.g., “I probably wouldn’t respond”). An independent-samples t -test on word count revealed a significant effect of chatbot message ($t(77) = 2.12$, $p = .04$). As expected, the warm chatbot message led participants to respond with a greater number of words ($M_{\text{warm}} = 10.23$ vs. $M_{\text{competent}} = 7.08$).

To further explore the nature of participants’ responses to the chatbot, we also analyzed how a warm (vs. competent) initial message affected four summary language variables that reflect important dimensions of the text responses. The four dimensions that are part of the LIWC output are analytical thinking, authenticity, clout, and emotional tone. These four summary variables’ scores range from 0 to 100 (Pennebaker, Boyd, et al., 2015). Considering brand–self distance as our proposed key mechanism, we were particularly interested in understanding if a warm (vs. competent) initial chatbot message would affect authenticity. Higher numbers on authenticity indicate that a response is sincere and personal rather than detached and distant. Accordingly, in recent research using LIWC, intimacy (i.e., closeness) has been associated with authenticity in language (Bloomfield et al., 2020; Li & Samp, 2019). Given that we predicted that a warm (vs. competent) initial message by a brand’s chatbot should make consumers feel closer to the brand, we expected that the warm (vs. competent) message would lead consumers to provide more authentic responses. We did not expect differences in the other dimensions, as they are not as closely tied to our theorization.

² Despite the limitations inherent in any convenience sample, MTurk has been validated as a reliable platform for the collection of academic research data. Empirical data suggest that MTurk samples tend to be of equal or better quality relative to both professional and student panels (Kees et al., 2017).

Table 1

Text analysis of consumer responses to a warm or competent initial chatbot message (Study 1).

Initial chatbot message	LIWC measures of consumer responses					Consumer response examples
	Word count	Authenticity	Emotional tone	Clout	Analytical thinking	
Warm	10.23 (7.29)	71.65 (31.87)	68.62 (36.22)	57.19 (36.82)	53.91 (35.48)	Hi! I have a few questions about my upcoming travels. What are the best places in the world to go to a beach? Once you slim that down, what time of year is the weather best to go? Thank you, I am looking to travel to Greece. Can you provide me with popular destinations? Hi, I would love to have a couple travel options first. Can I give you dates to look at? How much is a plane ticket to Greece in May? What is the best way to travel on a budget? I am curious how I can budget for travelling abroad as a college student. What places are safe to travel in Europe?
Competent	7.08 (5.85)	50.93 (45.03)	54.34 (36.92)	49.27 (28.68)	56.66 (38.09)	
	$p = .04$	$p = .02$	$p = .09$	$p = .29$	$p = .74$	

Notes: The table provides the means, with the standard deviations in parentheses. The p -values list the significance tests of the consumer responses to a warm (vs. competent) initial chatbot message for each LIWC measure. The examples represent the verbatim responses of the participants.

Specifically, higher scores on analytical thinking suggest logical and formal thinking, higher scores on clout suggest expertise, and higher scores on emotional tone reflect a more positive response.

Independent-samples t -tests on each of these summary variables confirmed our expectations. Participants who viewed the warm (vs. competent) chatbot message also provided a response that scored higher on authenticity ($M_{\text{warm}} = 71.65$ vs. $M_{\text{competent}} = 50.93$; $t(77) = 2.36$, $p = .02$). We found no significant differences on analytical thinking ($p = .74$) or clout ($p = .29$). However, the results did reveal a marginally significant effect of the warm (vs. competent) chatbot message on the respondents' emotional tone ($M_{\text{warm}} = 68.62$ vs. $M_{\text{competent}} = 54.34$; $t(77) = 1.74$, $p = .09$). While this was not our focal outcome, we believe that a warm (vs. competent) initial message may also improve how positively participants respond to the brand agent. Table 1 provides an overview of the LIWC results of Study 1.

4.3. Discussion

Study 1 provides initial evidence that a warm (vs. competent) message enhances consumer engagement with a brand's AI message. It also suggests that consumers feel closer to the brand after reading a warm (vs. competent) message, as evidenced by higher authenticity scores in their responses to the chatbot message. Therefore, this study indicates that a warm (vs. competent) initial message reduces brand–self distance. Our next study uses established scale measures to provide further empirical support for our proposed effects on brand–self distance and consumer engagement.

5. Study 2

Study 2 enhances the previous study in three main ways. First, we add a neutral condition to demonstrate that warm messages drive brand engagement. That is, the inclusion of a baseline or control condition allows us to demonstrate that a competent chatbot message does not reduce brand engagement; instead, a warm chatbot message increases consumer engagement. Second, Study 2 measures engagement using scale items that include the affective and cognitive dimensions of this construct (Brodie et al., 2011; Hollebeek, 2011). Third, to examine the underlying process, Study 2 measures brand–self distance as the proposed mediator.

5.1. Method

Two hundred twenty-six MTurk participants ($M_{\text{age}} = 37.50$ years; 53% female) were randomly assigned to the pretested warm, competent, or neutral initial text message by a chatbot operating on behalf of Viatti,

a fictitious travel brand. They then responded to a three-item scale measuring brand engagement (felt engaged with Viatti, motivated to learn more about Viatti, motivated to engage in the conversation further; 1 = not at all, 9 = very much; $\alpha = .96$; Kull & Heath, 2016; van Doorn et al., 2010). Thereafter, to measure brand–self distance, we adapted Park et al.'s (2013) two-item scale by asking participants how close and personally connected they felt to Viatti (1 = Viatti is very far away from me and who I am, 7 = Viatti is very close to me and who I am; 1 = I am personally disconnected from Viatti, 7 = I am personally connected to Viatti; $r = .93$). Finally, participants provided some general demographic information.

5.2. Results

An analysis of variance (ANOVA) on brand engagement revealed a significant effect of the tone of the chatbot's initial message ($F(2, 223) = 11.76$, $p < .001$, $\eta_p^2 = .10$). Specifically, planned contrasts revealed significant pairwise differences in that, as expected, a warm message ($M = 5.17$) resulted in greater engagement than a competent message ($M = 4.32$; $t(223) = 2.17$, $p = .03$) or a neutral message ($M = 3.30$; $t(223) = 4.84$, $p < .001$). Furthermore, the competent message induced greater engagement than the neutral message ($t(223) = 2.65$, $p = .009$).

An ANOVA on brand–self distance also revealed a significant effect of chatbot message ($F(2, 223) = 13.10$, $p < .001$, $\eta_p^2 = .11$). Planned contrasts showed the same pattern as the effects on engagement did. A warm message ($M = 3.81$) decreased perceived brand–self distance more (i.e., made consumers feel closer to the brand) than a competent message ($M = 3.07$; $t(223) = 2.92$, $p = .004$) or a neutral message ($M = 2.54$; $t(223) = 5.11$, $p < .001$) did, while the competent message made consumers feel closer to the brand than the neutral message did ($t(223) = 2.15$, $p = .03$).

Next, we tested our prediction that the effect of chatbot message on brand engagement would be mediated by brand–self distance. The independent variable was chatbot message, the dependent variable was brand engagement, and the mediator was brand–self distance. We indicated the multicategorical nature of the independent variable (warm, competent, or neutral message) when we ran the PROCESS macro (Hayes, 2018). Though PROCESS created two dummy variables, the focal independent variable was the dummy variable for the warm initial message. A mediation analysis using Model 4 from the PROCESS macro with 5,000 bootstrap samples revealed a significant effect of brand–self distance on engagement ($b = 1.21$, $SE = .07$, $t(222) = 18.30$, $p < .001$). Importantly, the results revealed a significant indirect effect of warm chatbot message on engagement through brand–self distance ($b = 1.53$, $SE = .30$, 95% confidence interval [CI] = [.96, 2.10]). With brand–self distance in the

model, the direct effect of warm chatbot message on engagement was no longer significant ($b = .34$, $SE = .26$, $t(222) = 1.31$, $p = .19$).³

5.3. Discussion

Using a realistic interaction scenario, our results show that a warm (vs. competent) initial chatbot message leads to higher consumer engagement. Furthermore, the results of Study 2 confirm that this enhanced engagement in the warm (vs. competent) message condition is explained by a reduction in brand–self distance. The inclusion of a neutral baseline condition allows us to disentangle the effects and validate that the proposed changes in engagement are mainly driven by warmth rather than competence. Specifically, it is the chatbot's warm message that increases consumer engagement with the brand as opposed to the chatbot's competent message that decreases it. In our next study, we provide further insight into our proposed brand–self distance account using moderation.

6. Study 3

In Studies 1 and 2, we have argued and demonstrated that a warm (vs. competent) chatbot message increases consumer engagement. Moreover, we have shown that this effect is driven by a reduction in brand–self distance. Our next study seeks to provide further evidence for this process through moderation of brand affiliation. That is, in Study 3, similar to previous research (Goodman & Lim, 2018), we manipulate social distance to a brand by asking participants to consider a brand with which they are affiliated or not affiliated.

6.1. Method

6.1.1. Pretests

For Study 3, we created two new chatbot messages, this time for the website of a bank. We pretested these messages to verify that one was indeed perceived as warm and the other one as competent. The chatbot's warm message read as follows: "How are you today? I am here to help you." Conversely, the chatbot's competent message read as follows: "Looking for something? I know all the products inside out." In addition to matching the general message style by having an initial question be followed by a brief self-description, we used a plain white background and also ensured consistency in word count across both conditions. Fifty MTurk participants ($M_{\text{age}} = 31.88$ years; 40% female) were randomly assigned to one of the two chatbot messages. They were then asked to answer a two-item bipolar scale that measured the extent to which the message came across as 1 = warm/friendly and 7 = competent/capable ($r = .90$). The manipulation worked as intended, as the warm message was perceived as warmer than the competent one ($M_{\text{warm}} = 3.96$ vs. $M_{\text{competent}} = 5.54$; $F(1, 48) = 11.85$, $p = .001$).

In Study 3, we used brand affiliation to manipulate rather than measure brand–self distance to create closeness or distance to a brand. Therefore, we also pretested that our manipulation was effective in creating the desired closeness or distance. We asked 61 MTurk respondents ($M_{\text{age}} = 35.79$ years; 41% female) to participate in a single-factor (brand affiliation: customer vs. noncustomer) between-subjects design. Half of the participants read that they were on the website of a bank with which they had been affiliated for years. The other half of the participants read that they were on the website of a bank with which they were not affiliated. After reading this scenario, they were asked to

answer two brand–self distance items (same as in Study 2; $r = .96$). As expected, when participants were told that they were affiliated with the bank, they reported a smaller brand–self distance ($M_{\text{customer}} = 5.20$) than when they were not affiliated with the bank ($M_{\text{noncustomer}} = 2.42$; $F(1, 59) = 58.52$, $p < .001$). Furthermore, a one-sample t -test revealed that the customer condition was significantly above the scale midpoint, indicating closeness ($t(29) = 4.72$, $p < .001$), whereas the noncustomer condition was significantly below the scale midpoint, indicating distance ($t(30) = -6.10$, $p < .001$).

6.1.2. Participants, design, and procedure

For Study 3, we recruited two hundred eighty-four MTurk participants ($M_{\text{age}} = 35.65$ years; 39% female). The study used a 2 (chatbot message: warm vs. competent) $\times$ 2 (brand affiliation: customer vs. noncustomer) between-subjects design. Participants were told that they were on the website of a bank with which they either had been affiliated for years (i.e., customer) or had no prior relationship (i.e., noncustomer). They were then randomly assigned to either the warm or the competent version of the pretested message of the bank's chatbot. Thereafter, participants responded to three brand engagement items on a 7-point scale (same as in Study 2; $\alpha = .93$). Lastly, they provided some general demographic information.

6.2. Results

We ran a two-way ANOVA testing the effects of chatbot message (warm vs. competent) and brand affiliation (customer vs. noncustomer) on engagement. The results revealed only a significant two-way interaction ($F(1, 280) = 9.00$, $p = .003$, $\eta_p^2 = .03$). No other main effect was significant ($ps > .27$). Planned contrasts revealed that when participants were affiliated with the brand, exposure to a warm (vs. competent) message led to higher engagement ($M_{\text{warm}} = 5.80$ vs. $M_{\text{competent}} = 4.72$; $F(1, 280) = 8.02$, $p = .005$, $\eta_p^2 = .03$). However, when participants were explicitly told that they were not affiliated with the brand, this difference between the two types of messages was attenuated ($M_{\text{warm}} = 5.15$ vs. $M_{\text{competent}} = 5.65$; $F(1, 280) = 1.90$, $p = .17$).

6.3. Discussion

Using a moderation-of-process approach, Study 3 provides supporting evidence for brand–self distance as the mechanism underlying the effect of a chatbot's warm (vs. competent) initial message on consumer engagement. In particular, consistent with our previous findings, we demonstrate in Study 3 that individuals exhibit higher engagement with a chatbot and its brand if the initial message is warm rather than competent. More specifically, we find that this tendency is exacerbated when we highlight to consumers that they have a close relationship with the brand. In such a context, we facilitate closeness to the brand, and a warm message enhances engagement. Conversely, when we emphasize to consumers that they have a distant relationship with the brand, we disrupt their development of a connection to the brand through its initial chatbot message, which in turn attenuates the effect on consumer engagement.

7. General discussion

Despite the growing frequency with which firms use chatbots as initial consumer touchpoints (e.g., Banfi, 2017; Verma, 2019), academic work exploring the role of chatbots in creating favorable impressions among consumers is lacking (e.g., Rafaeli et al., 2017; Robinson et al., 2020). The present research addresses this gap by systematically exploring how the either warm (i.e., friendly) or competent (i.e., capable) appeal of a chatbot's initial message can impact consumer responses. Specifically, three studies show that a warm (vs. competent) initial chatbot message fosters consumer–brand connections and ultimately engagement—unless the chatbot represents a brand that consumers have a purposefully distant

³ While not our focus, for completeness in reporting, we also note that the results revealed a significant indirect effect of a competent chatbot message on engagement through brand–self distance ($b = .64$, $SE = .31$, 95% CI = [.03, 1.23]). This suggests that, due to reduced brand–self distance, a competent message also increases engagement when compared with a neutral message, yet the effect is not as strong as that of a warm message.

relationship with, in which case the effect attenuates. These findings give rise to several implications for both theory and practice.

7.1. Theoretical contributions

Frontline technology must have the capability to engage consumers on a social level in order to contribute to a brand's success (van Doorn et al., 2017). According to social response theory, people tend to interact with computers as they would with other people (Holzwarth et al., 2006; Köhler et al., 2011). This adoption of social behaviors is facilitated by social cues, which are integral components of consumer reactions to brands in digital environments (Wang et al., 2017). We contribute to this emerging stream of research by showing that a warm (vs. competent) initial chatbot message is a sufficient social cue to increase consumer likelihood to respond. We also enhance social response theory by shedding light on the process that underlies consumer responsiveness to social cues in digital environments. Specifically, we argue and demonstrate that an initial warm (vs. competent) chatbot message decreases the perceived distance between the consumer and the brand that the chatbot represents, which in turn increases consumer response likelihood.

The current research enriches the growing literature that examines the evolving role of AI in marketing-related initiatives. Recent work suggests that the interaction style of smart products (i.e., Siri) can affect brand attachment (Wu et al., 2017) and, more generally, that consumers tend to prefer interacting with AI interfaces, including chatbots, that exhibit capabilities and characteristics that resemble humans (Huang & Rust, 2018; Shum et al., 2018). For example, consumers perceive humanlike (vs. cartoonlike) characters in e-commerce interfaces as more likeable, appropriate, and trustworthy (Luo et al., 2016) and are more likely to attempt to reconcile misunderstandings with humanlike (vs. obviously automated) chatbots (Corti & Gillespie, 2016). Similarly, providing a chatbot with humanlike features, such as a human name, has been found to enhance consumers' emotional connection to the chatbot's brand (Araujo, 2018). We contribute to this line of research by showing that beyond using the first person when conversing with consumers, any further humanization of chatbots, such as through visual cues or the provision of a name, is not a prerequisite for a stronger consumer–brand connection to occur. Importantly, we demonstrate that the warm (vs. competent) tone of the initial message of a brand's chatbot has a significant influence on consumers' likelihood to feel connected to the brand and, in turn, engage with it.

The present research enhances the SCM by providing further support for the robustness of warmth and competence as two fundamental dimensions of social cognition (Fiske et al., 2002, 2007). As shown across three experiments and associated pretests, general tendencies in impression formation of warmth and competence can be observed even after a mere exposure to an initial chatbot message. This novel finding further expands the scope of the SCM by highlighting the broad applicability and implications of the two universal dimensions of social cognition (Fiske et al., 2007). Specifically, we provide robust empirical evidence that a chatbot's warm (vs. competent) initial message brings consumers closer to the brand on whose behalf the chatbot operates, which in turn increases their motivations to learn more about the brand. This demonstrated superiority of warmth over competence also corroborates prior findings involving lexical decision tasks (Ybarra et al., 2001) and 100-ms face exposures (Willis & Todorov, 2006). Relatedly, we also enrich the BIAF (i.e., the extension of the SCM toward the branding domain; Kervyn et al., 2012) by providing empirical evidence for the development of brand perceptions and relationships on the basis of a warm (vs. competent) initial message of a brand's virtual service agent. Considering that we neither manipulated nor measured the personality of a brand as being warm or competent, our findings indicate that the warm or competent initial message of a chatbot can shape the personality of its brand to the extent that it can decrease consumers' perceived distance to the brand and foster brand engagement—even more so when the initial message comes across as warm rather than

competent. We document these effects not only through the use of engagement scales but also through qualitative consumer responses; as the results of Study 1 using a text analyzing tool have shown, consumers tend to reciprocate a chatbot's first endeavors to engage them in a conversation by responding more authentically when exposed to a chatbot's warm (vs. competent) initial message.

The current work addresses the need for more research that investigates the full spectrum of brand associations, including those with neutral and negative valence (Fournier & Alvarez, 2013; Park et al., 2013). While Studies 1 and 2 used fictitious brands, thus implicating a neutral brand association. Study 3 manipulated the perceived distance between consumers and a focal brand (i.e., brand–self distance). Respondents in one condition were told that they were previously affiliated with the focal brand, whereas those in the other condition were specifically told that they had no prior affiliation with the brand, thereby indicating a positive and negative brand association, respectively. Aside from providing further process evidence for brand–self distance as the mechanism underlying the warmth–engagement effect, Study 3 points to negative brand associations as an important boundary condition. More specifically, it appears that the favorable impact of a warm (vs. competent) initial chatbot message on consumer engagement is restricted to brands that consumers either do not know or have a positive preexisting relationship with. For brands that consumers have already chosen not to do business with, coming across as warm and competent seems equally important to foster consumer engagement. This differential effect supports the notion that there is yet much to learn about the implications of distant brand relationships, which may not need to be aversive to depart from how neutral or positive brand relationships can be leveraged (Fournier & Alvarez, 2013).

With the present research, we respond to recent calls for research on how AI platforms, such as chatbots, can be employed as facilitators of consumer engagement (Kumar et al., 2021; Marketing Science Institute, 2020). Similarly, highlighting the important role of AI in personalized engagement marketing, Kumar et al. (2019) propose that brand awareness and preexisting brand trust should assist AI tools, such as chatbots, in fostering engagement, as they tend to require consumers to make a conscious effort to learn about the brand's offerings. Study 3 finds support for this proposition by reporting an accentuation of the warmth–engagement effect for preexisting positive brand relationships. At the same time, however, Study 3 highlights the need for caution in that brand awareness alone may also hinder AI technology in the event that the preexisting brand perception is not favorable. More generally, we join growing efforts to shed light on the antecedents of consumer engagement (e.g., Hollebeek, 2011; Pansari & Kumar, 2017; van Doorn et al., 2010). Our findings confirm both warmth as an important driver in connecting consumers to brands (Kolbl et al., 2019; Stokburger-Sauer et al., 2012) and brand connectedness or closeness as an important driver in fostering engagement (Hsieh & Chang, 2016; Pansari & Kumar, 2017). The present work also corroborates the tone of a message as an essential content characteristic in brand communications. While prior research has found that message tone impacts a brand's persuasion effectiveness in social media (Stephen et al., 2015), we show that message tone influences consumers' response likelihood to a brand's initial chatbot message.

7.2. Managerial implications

Results from three experimental studies indicate that, in general, managers who seek to foster consumer–brand relationships should consider employing chatbots with warm (vs. competent) initial messages. This insight is particularly actionable information for the many startup firms who face the challenging task of establishing relationships with consumers who are often already loyal to other brands. Our findings suggest that a warm (vs. competent) initial message decreases consumers' perceived distance to the unknown brand, which in turn increases their motivation to learn more about the brand—a window of opportunity for new brands to make an immediate strong impression on prospective customers. Thus, chatbots offer managers an

opportunity to not only cut operational costs (e.g., [Canhoto & Clear, 2020](#); [Jovic, 2020](#)) but also efficiently capture and convert prospects to customers.

Our findings indicate that established brands would similarly benefit from initiating online conversations with existing consumers using warm chatbot messages. Such customers already have close brand relationships and thus tend to be the brand's most valuable segment. Managers would thus be well advised to enter into virtual conversations with this segment using warm messages, as these customers tend to implicitly expect the closeness of the existing relationship, in which they have already invested, to be acknowledged and reciprocated in the chatbot's initial message delivery.

However, the current work also shows that having a chatbot send a warm initial message is not a one-size-fits-all strategy. As Study 3 reveals, the warmth–engagement effect attenuates for brands that consumers have chosen not to be affiliated with (i.e., feel distant toward). Hence, managers of such brands should thoughtfully consider the context and the reasons for the deterioration or lack of consumer–brand relationships in order to determine whether a chatbot's warm or competent first impression might increase consumers' likelihood of initially interacting with the chatbot and ultimately engaging with the brand it serves.

An additional insight is related not only to consumers' willingness to engage but also to their type of engagement. Specifically, the results of Study 1 indicate that when chatbots approach a conversation using warmth, consumers respond more authentically. This finding provides managers with a heightened opportunity for furthering the consumer–brand relationship and also for gaining additional insight into consumers' motivations and mindsets.

The results from our three experiments serve as a reminder that chatbots must be viewed as not only an AI-powered software tool but also a critical opportunity for initiating and managing consumer relationships. Brands should allocate resources accordingly to ensure that the chatbot's initial tone meets customer expectations. Our findings are particularly significant given the weight an initial impression carries in an online context. Consumers must be attracted and engaged quickly, as most users decide whether to leave a website within just one minute ([Nielsen, 2011](#)). While chatbots can help gauge consumer interest, a cold initial chatbot message, which discourages consumers from responding, risks unraveling all technological investments in a chatbot's rigorous request-handling capabilities. The strategic implementation of this technology requires an implicit understanding that chatbots not only represent the brand, thereby providing opportunities for branded service encounters ([Sirianni et al., 2013](#)), but most often also serve as the initial touchpoint in consumer–brand interactions. Overall, our findings consistently show that both unknown and liked brands can benefit from having their chatbots send warm and friendly initial messages to bring consumers closer, which in turn motivates consumers to spend more time getting to know the brands.

7.3. Limitations and further research

Drawing on the SCM and research on general impression formation (e.g., [Fiske et al., 2002](#); [Wojciszke et al., 1998](#)), the current work reveals that, for most brands, a warm (vs. competent) initial chatbot message generates more favorable consumer brand perceptions and behaviors. Though beyond the scope of the present research, future studies should explore whether and how consumer responses to message warmth or competence tend to alter as the interactions continue. Is the superiority of good intentions (i.e., warmth) over the ability to act upon these good intentions (i.e., competence) restricted to the initial impression, and if so, do the effects reverse as the conversation unfolds? Moreover, are the longitudinal effects of warmth and competence influenced by the nature

and strength of the preexisting consumer–brand relationship? Additional research is warranted to explore these theoretically intriguing and managerially relevant questions.

To specifically examine the role of a chatbot's initial message tone in affecting brand outcomes, we controlled for a range of other potential cues that may influence consumer responses. Previous research has indicated that the humanization of service agents through anthropomorphic cues can benefit brand outcomes. For example, using avatars as sales agents can enhance brand satisfaction, product attitudes, and purchase intentions ([Holzwarth et al., 2006](#)). Similarly, giving a chatbot a name can enhance consumers' perceived connection to the chatbot's brand ([Araujo, 2018](#)). Another visual cue in this context is the typeface that brands choose for chatbot interactions, which can impact the extent to which consumers believe they are interacting with a human or a machine ([Candello et al., 2017](#)). In a broader context (e.g., advertising, product packaging, websites), handwritten typefaces have been found to increase consumer perceptions of human presence ([Schroll et al., 2018](#)). Therefore, it would be interesting to explore whether and how these and other visual and humanlike cues interact with the tone (i.e., warmth or competence) of a chatbot's initial message. For instance, would a chatbot's warm or competent initial message benefit more from using an avatar, chatbot name, or handwritten typeface? And if so, how?

Finally, the present research provides promising avenues for further investigation through the consideration of cultural, individual, and situational factors. For example, based on cross-cultural research ([Hofstede et al., 2010](#)), it seems plausible that the warmth effect on engagement may reverse, or at least attenuate, in cultures that score low in indulgence or high in restraint, such as China ([Hofstede Insights, 2021](#)). An individual difference variable that may affect the results is one's need for cognition (i.e., the tendency to engage in and enjoy effortful thinking; [Cacioppo & Petty, 1982](#)). For consumers with low (high) need for cognition, the warmth–engagement effect may accentuate (attenuate). An example for a situational factor that may influence the results could be a temporary sense of loneliness. Because an elevated degree of loneliness makes people more sensitive to cues that implicate opportunities for the formation of social relationships ([DeWall et al., 2009](#)) and makes them more inclined to humanize nonhuman agents ([Wan & Aggarwal, 2015](#)), our observed effects may accentuate for consumers high in perceived social disconnection or loneliness. Specifically, they may feel more quickly connected to a brand whose chatbot initiates an interaction with a warm message and be more motivated to engage in the conversation further than those who feel less disconnected or lonely at a given time.

8. Conclusion

Despite the ever-increasing extent to which brands rely on virtual service agents in the form of chatbots on their websites and mobile applications, the branding implications of such practices have received limited academic attention. Three studies exploring the first impression of chatbots on consumers' brand perceptions and responses converge on the conclusion that a warm (vs. competent) initial message makes consumers feel closer to the brand, which in turn increases consumers' likelihood to respond to the chatbot and further engage with the brand. However, our empirical findings reveal that these effects are not necessarily universal. Specifically, the influence of a warm (vs. competent) initial chatbot message is moderated by brand affiliation, such that the effect attenuates when consumers are explicitly told that they have no prior relationship with the brand represented by the chatbot. Taken together, these insights provide promising avenues for further research in this emerging area of AI chatbots and their implications for brands and brand relationships.

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